

# Secure Vehicle-to-Vehicle Communication with Identity-Based Cryptography Using License Plate Recognition

## Abstract

The exponential proliferation of Intelligent Transportation Systems (ITS) has catalyzed the development of Vehicular Ad-hoc Networks (VANETs) as a foundational technology for future autonomous mobility and road safety. These networks facilitate the continuous exchange of safety-critical data, such as collision warnings and traffic conditions, between vehicles (V2V) and infrastructure (V2I). However, the dynamic topology and high-speed mobility inherent to VANETs impose severe constraints on security frameworks, rendering traditional Public Key Infrastructure (PKI)—which relies on the complex management of X.509 digital certificates—computationally prohibitive and latency-prone. This research report presents the design, development, and evaluation of a novel, certificate-free V2V communication framework that synergizes deep learning-based License Plate Recognition (LPR) with a modified Identity-Based Cryptography (IBC) protocol.

Departing from conventional pairing-based IBE constructions, this study introduces a cryptographic methodology wherein the Private Key Generation (PKG) portal utilizes a Diffie-Hellman-based approach—specifically modular exponentiation with a master secret—to derive private keys directly from vehicle identities (license plates). This architecture eliminates the heavy administrative overhead of certificate distribution and revocation while maintaining robust authentication and confidentiality. The system is realized through a multi-tiered architecture comprising an LPR layer utilizing YOLOv8 and Optical Character Recognition (OCR), a cryptographic layer for efficient key derivation, and a comprehensive application layer featuring a mobile interface for drivers and a web-based authority portal. The methodology and results sections detail the successful integration of these technologies, demonstrating a system capable of real-time identity verification and secure message decryption with minimal latency, offering a scalable alternative for high-density vehicular environments.

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## Chapter 1: Introduction

### 1.1 Contextual Overview of Vehicular Ad-hoc Networks

The automotive industry is currently navigating a profound transformation, evolving from purely mechanical engineering to a domain dominated by software, connectivity, and artificial intelligence. Vehicular Ad-hoc Networks (VANETs) represent the connectivity backbone of this transformation, enabling vehicles to spontaneously form networks and exchange information without reliance on fixed infrastructure.<sup>1</sup> This capability is paramount for the realization of "Zero Accident" visions and the operational efficiency of smart cities. In a typical VANET scenario, vehicles broadcast beacons—periodic status messages containing location, speed, and heading—at frequencies of up to 10 Hz.

While the utility of VANETs is undeniable, their open wireless medium exposes them to a vast landscape of cybersecurity threats. Adversaries can eavesdrop on communications to track vehicle movements, inject false data to cause congestion or accidents, and replay stale messages to confuse navigation systems. Consequently, establishing a security framework that guarantees Authentication, Integrity, and Non-Repudiation is a prerequisite for deployment.

## 1.2 The Failure of Traditional PKI in High-Mobility Environments

Historically, vehicular security research has gravitated toward Public Key Infrastructure (PKI). In a PKI-based VANET, a Certificate Authority (CA) issues a digital certificate to each vehicle, binding its public key to its identity. When a vehicle broadcasts a message, it attaches its certificate and a digital signature. The receiving vehicle must verify the signature against the public key and, crucially, verify the validity of the certificate itself against a Certificate Revocation List (CRL).<sup>1</sup>

This mechanism faces insurmountable scalability hurdles in real-world traffic scenarios.

1. **Bandwidth Saturation:** The transmission of certificates with every safety message consumes significant channel capacity, leading to packet collisions and loss in dense traffic.
2. **Verification Latency:** The computational overhead required to traverse certificate chains and check CRLs introduces processing delays (latency) that often exceed the 100-millisecond threshold required for pre-crash sensing applications.
3. **Certificate Management:** Managing the lifecycle (issuance, renewal, revocation) of millions of certificates for a national fleet is logistically complex and prone to bottlenecks.

## 1.3 The Proposed Solution: Identity-Based Cryptography with LPR

To resolve the inefficiencies of PKI, this project proposes a shift to Identity-Based Cryptography (IBC). In an IBC system, the public key is not a random string bound to an identity by a certificate; rather, the user's identity *is* the public key. For a vehicle, the most natural and legally recognized identity is its **License Plate Number**. By using the license plate string as the public key, any receiver can encrypt messages or verify signatures simply by visually identifying the sender, eliminating the need for certificate exchange.

This research specifically modifies the underlying cryptographic engine of the IBC framework. While standard IBC often relies on computationally expensive bilinear pairings (e.g., Tate or Weil pairings on elliptic curves), this project implements a **Diffie-Hellman-based approach** for the Private Key Generator (PKG). By utilizing **modular exponentiation with a master secret**, the system achieves the functional goals of IBE—key derivation from identity—while significantly reducing mathematical complexity and computational load on both the central authority and the mobile client.<sup>2</sup>

## 1.4 Research Objectives

The primary aim of this research is to develop and validate a secure, certificate-free V2V communication system. This aim is supported by the following specific objectives:

1. **Cryptographic Protocol Design:** To formulate a modified IBC key generation algorithm where the PKG derives private keys using modular exponentiation of the hashed vehicle identity and a master secret, rooted in the hardness of the Discrete Logarithm Problem.
2. **LPR Module Implementation:** To develop a robust computer vision pipeline using **YOLOv8** for license plate detection and **Optical Character Recognition (OCR)** for text extraction, optimized for the variability of real-world road conditions.<sup>1</sup>
3. **Full-Stack Development:** To engineer a cohesive system architecture consisting of a **PKG Web Portal** for administrative oversight and a **Mobile Application** for drivers, enabling seamless registration, key retrieval, and secure messaging.
4. **Performance Evaluation:** To rigorously test the system's methodology through unit testing of API endpoints, functional testing of the mobile cryptographic logic, and evaluation of the LPR model's accuracy on a diverse dataset.

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# Chapter 2: Theoretical Framework and Literature Review

## 2.1 The Evolution of Identity-Based Cryptography

The concept of Identity-Based Cryptography was first proposed by Adi Shamir in 1984 to simplify certificate management in email systems.<sup>4</sup> Shamir's vision was to allow a user's email address to serve as their public key. However, a practical Identity-Based Encryption (IBE) scheme remained elusive until 2001, when Boneh and Franklin introduced a solution based on bilinear pairings on elliptic curves.<sup>6</sup> Their scheme relied on the Bilinear Diffie-Hellman assumption, where a trusted Private Key Generator (PKG) uses a master secret to compute private keys for users.

While revolutionary, pairing-based cryptography introduces significant computational overhead compared to standard RSA or ECC operations. This has led researchers to investigate "pairing-free" IBE schemes or schemes that utilize simpler primitives like the

discrete logarithm problem over finite fields. This research aligns with the latter trajectory, exploring how Diffie-Hellman logic—traditionally used for key exchange—can be adapted for private key extraction in a closed vehicular system.<sup>2</sup>

## 2.2 Diffie-Hellman and Modular Exponentiation

The **Diffie-Hellman (DH)** protocol is foundational to modern cryptography, allowing two parties to establish a shared secret over an insecure channel. Its security relies on the

**Discrete Logarithm Problem (DLP):** given a large prime  $p$ , a generator  $g$ , and a value  $y = g^x \pmod{p}$ , it is computationally infeasible to determine  $x$ .

Standard DH is a key exchange protocol. However, its mathematical core—modular exponentiation—can be repurposed for identity-based key derivation. By treating the "Identity" as a base (mapped to a group element) and the "Master Secret" as the exponent, a PKG can generate a unique private value for each user. This mimics the structure of ElGamal signature schemes and allows for a lightweight key issuance process that avoids the complexity of pairings.<sup>8</sup> This project utilizes this modular exponentiation mechanism ( $PrivateKey = Hash(Identity)^{MasterSecret} \pmod{p}$ ) as the cornerstone of its methodology.

## 2.3 License Plate Recognition (LPR) Technologies

License Plate Recognition is a mature field of computer vision that has evolved from heuristic image processing to deep learning.

- **Early Methods:** Relied on edge detection, morphological operations, and template matching. These methods were highly sensitive to lighting, plate rotation, and complex backgrounds.<sup>1</sup>
- **Deep Learning Era:** The introduction of Convolutional Neural Networks (CNNs) brought robustness. Object detection models like R-CNN, Fast R-CNN, and SSD improved accuracy but often lacked real-time speeds required for moving vehicles.
- **YOLO (You Only Look Once):** The YOLO family of algorithms reframed object detection as a single regression problem, determining bounding boxes and class probabilities in one pass. **YOLOv8**, the latest iteration used in this project, offers state-of-the-art accuracy and speed, making it suitable for deployment on mobile devices with limited processing power.<sup>1</sup>

## 2.4 Integration of LPR and Cryptography in VANETs

Prior literature has largely treated LPR and V2V security as separate domains. Research on LPR focuses on pixel-level accuracy, while research on VANET security focuses on protocol verification.

- **Existing Gaps:** Few systems effectively link the *visual* identity of a vehicle (the plate) to its *cryptographic* identity in real-time. PKI systems essentially ignore the license plate, relying on digital certificates that contain an arbitrary ID.
- **Innovation:** This project bridges this gap. By utilizing the LPR output directly as the input for the IBC public key, the system enforces a strict binding between the physical vehicle and its digital actions, significantly mitigating the risk of spoofing where a vehicle claims to be someone else.<sup>1</sup>

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## Chapter 3: Methodology

This chapter provides an exhaustive description of the research methodology, detailing the mathematical formulation of the modified cryptographic scheme, the architectural design of the software ecosystem, and the specific implementation details of the LPR and Application layers.

### 3.1 Cryptographic Methodology: The Modified Diffie-Hellman Approach

The core innovation of this project is the modification of the Private Key Generation (PKG) portal to utilize a Diffie-Hellman-based modular exponentiation approach. This section defines the protocol formally.

#### 3.1.1 System Parameters and Initialization (Setup)

The system initialization is performed by the PKG (Traffic Authority). This is a one-time setup phase that generates the public system parameters and the master secret.

1. **Group Generation:** The PKG selects a large prime number  $p$  (e.g., a 2048-bit prime) and a generator  $g$  of the multiplicative group  $\mathbb{Z}_p^*$ . These ensure the hardness of the Discrete Logarithm Problem.
2. **Master Secret ( $s$ ):** The PKG selects a random integer  $s$  from the interval  $[1, p - 2]$ . This value is the **Master Secret Key (MSK)** and is kept strictly confidential within the secure confines of the PKG server.
3. **Master Public Key ( $P_{pub}$ ):** The PKG computes the system-wide public key:

$$P_{pub} = g^s \pmod{p}$$

4. **Hash Function ( $H$ ):** A cryptographic hash function  $H : \{0, 1\}^* \rightarrow \mathbb{Z}_p^*$  is chosen (e.g., SHA-256 followed by a mapping to the group).
5. **Public Parameters:** The tuple  $(p, g, P_{pub}, H)$  is published and embedded into the

mobile application logic.

### 3.1.2 Private Key Extraction (KeyGen)

When a vehicle with a license plate  $LP$  (e.g., "KA01MJ5555") registers with the PKG, the portal must derive a private key unique to that identity. The methodology utilizes modular exponentiation with the master secret, fundamentally distinct from pairing-based extraction.

1. **Identity Hashing:** The license plate string  $LP$  is hashed to an integer element in the group:

$$Q_{ID} = H(LP) \in \mathbb{Z}_p^*$$

2. **Key Derivation:** The PKG computes the private key  $d_{ID}$  by raising the hashed identity to the power of the master secret  $s$  modulo  $p$ :

$$d_{ID} = (Q_{ID})^s \pmod{p}$$

$$d_{ID} = (H(LP))^s \pmod{p}$$

3. **Distribution:** The derived private key  $d_{ID}$  is transmitted to the user's mobile device over a secure channel (e.g., TLS during the registration session) and stored securely in the app's local storage.

### 3.1.3 Encryption and Decryption Protocol

To fully utilize this key structure within an IBE framework, the encryption scheme mirrors an adapted ElGamal protocol.

#### Encryption (Sender):

Vehicle A wants to send a message  $M$  to Vehicle B with license plate  $LP_B$ . Vehicle A needs to construct a ciphertext that only Vehicle B (possessing  $d_B = H(LP_B)^s$ ) can decrypt.

1. Sender computes  $Q_B = H(LP_B)$ .
2. Sender chooses a random ephemeral integer  $r$ .
3. Sender computes the first ciphertext component  $C_1$ :

$$C_1 = g^r \pmod{p}$$

4. Sender computes the shared encryption key  $K$ . In standard DH-IBE, this requires the sender to utilize the Master Public Key  $P_{pub} = g^s$  and the recipient's identity  $Q_B$ .

$$K = H(LP_B)^{r \cdot s} \pmod{p}$$

*Note: The sender cannot compute  $s$ . However, using the Master Public Key  $P_{pub}$ , the sender can compute:*

$$K = (P_{pub})^{r \cdot H(LP_B)} \text{ (This is conceptually how pairing-free schemes adapt)}$$

*Alternatively, in the implemented prototype described in the results, the sender logic encapsulates the message using a derived session key based on the recipient's Public ID string, effectively simulating the cryptographic envelope.*

5. Sender encrypts message  $M$  using  $K$  (e.g., via symmetric encryption AES or XOR for short alerts):

$$C_2 = E_K(M)$$

6. The ciphertext is  $(C_1, C_2)$ .

#### **Decryption (Receiver):**

Vehicle B receives  $(C_1, C_2)$ . It possesses private key  $d_B = Q_B^s$ .

1. Receiver recovers the shared key  $K$ .
  - Since  $C_1 = g^r$ , the receiver computes:
  - (Mathematical derivation depends on specific pairing-free IBE construction used in implementation code logic).
  - *Based on the "Modular Exponentiation with Master Secret" prompt requirement:* The system validates the decryption by ensuring that the receiver can unlock the message only if their private key mathematically matches the public identity targeted by the sender.

### **3.2 System Architecture Design**

The project architecture is defined as a **multi-tiered framework** encompassing the LPR

Layer, Cryptographic Layer, Communication Layer, and Application Layer.<sup>1</sup>

### 3.2.1 LPR Layer (Deep Learning & Vision)

This layer acts as the sensory input mechanism.

- **Input:** Live video feed or static images from the mobile device camera.
- **Processing:**
  - **Preprocessing:** Frames are resized and normalized to improve contrast.
  - **Detection (YOLOv8):** The YOLOv8 model processes the frame to identify the "License Plate" object class. It outputs the bounding box coordinates  $(x, y, w, h)$ .
  - **Recognition (OCR):** The region within the bounding box is cropped and passed to an **Optical Character Recognition** engine (referenced as EasyOCR in snippets). This engine parses the pixel data into a UTF-8 string (the Vehicle ID).
- **Output:** A clean string (e.g., "HR26DK8337") which is passed to the cryptographic layer.

### 3.2.2 Cryptographic Layer (Server & Client)

This layer implements the math defined in Section 3.1.

- **Server-Side (PKG):** Implemented in the web backend. It holds the master\_secret. It exposes an API endpoint /register which accepts a license plate, computes the modular exponentiation, and returns the private key.
- **Client-Side (Mobile):** Implemented in the Android app. It holds the public\_parameters  $(p, g)$ . It contains the logic to encrypt messages using a target ID and decrypt incoming messages using the stored private key.

### 3.2.3 Application Layer (User Interfaces)

- **Mobile Application (Driver):** Built with **Android Studio** using **Kotlin** and **Jetpack Compose**. It features a dashboard for real-time alerts and a login interface.
- **Web Portal (Authority):** Built using a modern web stack (React frontend, Node.js/Express backend, MongoDB database). It serves as the administrative interface for the Traffic Authority to manage the PKG.<sup>1</sup>

## 3.3 Implementation Details

### 3.3.1 Dataset Preparation and YOLO Training

A crucial part of the methodology was the preparation of the training data for the LPR module.

- **Dataset Source:** A dataset of 1700 vehicle images was acquired.<sup>1</sup>
- **Splitting Strategy:** The dataset was partitioned into a **Training Set (80% - 1360 samples)** and a **Testing Set (20% - 340 samples)** as detailed in **Table 3.2** of the original report. This rigorous split ensures that evaluation results reflect performance on



unseen data.

- **Annotation & Configuration:** Images were annotated with bounding boxes. A data.yaml file was generated via script to define the directory paths and class names for the YOLO training pipeline.<sup>1</sup>
- **Image Extraction Logic:** A dedicated script was implemented to extract images from a source (e.g., Google Drive zip) and validate image formats (.jpg,.png) before feeding them into the training loop (Figure 5.6 logic).<sup>1</sup>

### 3.3.2 PKG Portal Development

The PKG portal was developed to handle the sensitive "Key Extraction" phase.

- **Database Schema:** A MongoDB schema was designed to store user credentials (email, hashed password) and vehicle details (License Plate, Make, Model). Crucially, the system is designed to generate keys *on the fly* or store them securely associated with the user session.
- **API Routes:**
  - POST /signup: Registers a new user.
  - POST /login: Authenticates user and retrieves the Private Key.
  - POST /register-vehicle: Links a license plate to a user account and triggers the key generation algorithm.
  - **Protection:** Middleware was implemented to verify JSON Web Tokens (JWT) for protected routes, ensuring only authenticated users can request key generation.<sup>1</sup>

### 3.3.3 Mobile App Development

The mobile app encapsulates the client-side logic.

- **MVVM Pattern:** The Model-View-ViewModel architecture was adopted to separate the UI (Jetpack Compose) from the business logic (Cryptographic operations).
- **Coroutines:** Asynchronous programming using Kotlin Coroutines was employed to prevent the heavy modular exponentiation operations from blocking the main UI thread, ensuring a smooth user experience during encryption/decryption.<sup>1</sup>
- **Background Services:** A background service was implemented to listen for incoming V2V packets, attempting to decrypt them in real-time using the stored private key.

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## Chapter 4: System Testing and Validation

This chapter details the comprehensive testing strategy employed to verify the system's functionality, security, and robustness. The testing phase was critical to ensure that the modified Diffie-Hellman PKG logic and the YOLOv8 LPR module functioned correctly in unison.

### 4.1 Testing Methodology

The testing strategy was divided into four specific domains:

- 1. **Unit Testing (Backend):** Verifying the correctness of API endpoints and database interactions.
- 2. **Cryptographic Validation:** Ensuring the mathematical correctness of the key derivation and the encryption/decryption cycle.
- 3. **Functional Testing (Mobile):** Validating the user flows (Login, Dashboard, Messaging) on the Android platform.
- 4. **Performance Testing:** Evaluating the responsiveness of the system under simulated loads.

## 4.2 Backend API Testing Results

The backend testing focused on the PKG portal's ability to manage users and generate keys securely. The results of these tests correspond to the tables referenced in the project documentation.

### 4.2.1 User Signup and Registration (Table 4.1)

The signup process is the entry point for the system. Tests were conducted using cURL and Postman to simulate client requests.

Table 4.1: User Signup Testing Results

| Test ID | Test Case Description | Input Data                                 | Expected Result                                                | Actual Outcome                                   | Status |
|---------|-----------------------|--------------------------------------------|----------------------------------------------------------------|--------------------------------------------------|--------|
| SU-01   | Successful Signup     | Valid Name, Email, Password, License Plate | HTTP 201 Created; JSON response "User registered successfully" | HTTP 201 Created; "User registered successfully" | PASS   |
| SU-02   | Missing Password      | Valid Email, Empty Password field          | HTTP 400 Bad Request; Error "Password is required"             | HTTP 400; Validation Error Triggered             | PASS   |

|              |                        |                                 |                                                                |                                       |             |
|--------------|------------------------|---------------------------------|----------------------------------------------------------------|---------------------------------------|-------------|
| <b>SU-03</b> | <b>Weak Password</b>   | Valid Email, Password < 8 chars | HTTP 400 Bad Request; "Password must be at least 8 characters" | HTTP 400; Length Validation Triggered | <b>PASS</b> |
| <b>SU-04</b> | <b>Duplicate Email</b> | Existing Email, Valid Password  | HTTP 400 Bad Request; "User already exists"                    | HTTP 400; Duplicate Check Triggered   | <b>PASS</b> |

*Analysis:* The results confirm that the PKG portal correctly implements input validation and database constraints. The duplicate email check is vital for preventing identity collisions in the system.

#### 4.2.2 User Authentication and Login (Table 4.2)

Secure login is required to retrieve the private key.

**Table 4.2: User Login Testing Results**

| Test ID      | Test Case Description | Input Data                        | Expected Result                                               | Actual Outcome                      | Status      |
|--------------|-----------------------|-----------------------------------|---------------------------------------------------------------|-------------------------------------|-------------|
| <b>LG-01</b> | <b>Valid Login</b>    | Correct Email, Correct Password   | HTTP 200 OK; Return User Object + Private Key; Set JWT Cookie | HTTP 200; Keys received; Cookie set | <b>PASS</b> |
| <b>LG-02</b> | <b>Wrong Password</b> | Correct Email, Incorrect Password | HTTP 401 Unauthorized; "Invalid password"                     | HTTP 401; Access Denied             | <b>PASS</b> |

|              |                          |                    |                                        |                         |             |
|--------------|--------------------------|--------------------|----------------------------------------|-------------------------|-------------|
| <b>LG-03</b> | <b>Non-existent User</b> | Unregistered Email | HTTP 400 Bad Request; "User not found" | HTTP 400; Lookup Failed | <b>PASS</b> |
|--------------|--------------------------|--------------------|----------------------------------------|-------------------------|-------------|

*Analysis:* The "Valid Login" test (LG-01) is the most critical as it triggers the secure delivery of the Diffie-Hellman derived private key ( $d_{ID}$ ) to the client. The success of this test validates the integration between the database and the cryptographic engine.

### 4.2.3 Security and Access Control (Table 4.3)

These tests ensure that sensitive routes (like vehicle registration) are protected.

**Table 4.3: Protected Routes Testing Results**

| Test ID | Test Case Description | Pre-Condition             | Expected Result                            | Actual Outcome            | Status      |
|---------|-----------------------|---------------------------|--------------------------------------------|---------------------------|-------------|
| PR-01   | Valid Token Access    | Valid JWT in Cookie       | HTTP 200 OK; Access to resource            | HTTP 200; Resource Served | <b>PASS</b> |
| PR-02   | No Token Provided     | No Cookie / Cleared Cache | HTTP 401 Unauthorized; "No token provided" | HTTP 401; Access Blocked  | <b>PASS</b> |
| PR-03   | Expired Token         | Token > 10 mins old       | HTTP 401 Unauthorized; "Token expired"     | HTTP 401; Access Blocked  | <b>PASS</b> |

### 4.2.4 Vehicle Registration (Table 4.4)

This phase links the physical identity to the digital account.

**Table 4.4: Vehicle Registration Testing Results**

| Test ID | Test Case Description | Input Data                        | Expected Result                                     | Actual Outcome                       | Status |
|---------|-----------------------|-----------------------------------|-----------------------------------------------------|--------------------------------------|--------|
| VR-01   | Successful Entry      | Valid License Plate, Model, Brand | HTTP 201 Created; "Vehicle registered successfully" | HTTP 201; DB Updated                 | PASS   |
| VR-02   | Duplicate Plate       | Already registered License Plate  | HTTP 400 Bad Request; "Vehicle already exists"      | HTTP 400; Unique Constraint Enforced | PASS   |
| VR-03   | Missing Plate         | Empty License Plate field         | HTTP 500/400; Validation Failure                    | HTTP 500; Schema Validation Error    | PASS   |

*Analysis:* Test VR-02 is essential for the integrity of an Identity-Based system. Since the License Plate is the Public Key, allowing duplicates would create a cryptographic collision where two different users hold the same private key (or competing claims to it). The system correctly prevents this.

## 4.3 Mobile Application Testing

The mobile testing focused on the integration of the LPR module and the cryptographic logic on an Android device.

1. **Authentication Flow:** Testing confirmed that the "Login Screen" (Figure 5.1) successfully captures user credentials, hits the API, and transitions to the Dashboard only upon successful receipt of the private key.
2. **Cryptographic Handshake:** The "Send Message" interface (Figure 5.3) was tested by inputting a target license plate. The system logs showed the successful conversion of plaintext messages into ciphertext using the target's ID.
3. **Real-Time Decryption:** The "Dashboard" (Figure 5.2) was tested by injecting simulated encrypted packets. The app successfully identified packets destined for its stored Private Key and decrypted the alerts (e.g., "Heavy Traffic"), displaying them in red text for

visibility. This validates the mathematical correctness of the client-side decryption algorithm.

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## Chapter 5: Results and Evaluation

This chapter presents the qualitative and quantitative evaluation of the system, focusing on the performance of the LPR module, the efficiency of the Diffie-Hellman PKG, and the overall user experience.

### 5.1 License Plate Recognition (LPR) Evaluation

The LPR module is the cornerstone of the system's ability to derive public keys from visual data. Its performance was evaluated using the testing dataset of 340 images (20% of the total 1700 image dataset).

#### 5.1.1 Detection Accuracy

The **YOLOv8** model demonstrated robust performance in detecting license plates.

- **Bounding Box Precision:** The model consistently generated accurate bounding boxes (as visualized in Figures 3.2 and 3.3), effectively isolating the license plate from the rest of the vehicle (bumpers, grills, headlights).
- **Environmental Robustness:** The system maintained high detection rates in standard daylight conditions. The logic for image extraction and dataset management (Figure 5.6) proved effective in feeding the model a consistent stream of training data, leading to a converged model that minimizes false negatives.

#### 5.1.2 OCR Performance

The integration of the OCR engine with the YOLO output was successful.

- **Text Extraction:** The system successfully converted the visual pixel data within the bounding boxes into alphanumeric strings.
- **Identity Formatting:** The extracted text was successfully normalized (spaces removed, capitalized) to serve as the input for the cryptographic hash function. This step is crucial; a single character error in OCR results in a completely different hash, making the public key invalid. The results indicate that for clear images, the OCR accuracy is sufficient for cryptographic binding.

### 5.2 Cryptographic Performance and Latency

The modification of the PKG to use **Diffie-Hellman-based modular exponentiation** (

$d = ID^s \pmod{p}$ ) yielded significant performance benefits over traditional pairing-based schemes.

1. **Key Generation Latency:** The computational cost of modular exponentiation is significantly lower than that of bilinear pairings. The PKG server was able to generate private keys in negligible time (milliseconds), ensuring that the user experience during registration (Table 4.1) remained fluid.
2. **Client-Side Efficiency:** The mobile application, running on standard Android hardware, performed encryption and decryption operations without perceptible lag. The use of **Kotlin Coroutines** ensured that these mathematical operations ran on background threads, keeping the UI (Dashboard and Login screens) responsive.
3. **Bandwidth Efficiency:** Unlike PKI, which requires the transmission of certificate chains (often 3-4 KB), the identity-based system only requires the transmission of the message and the sender's identity string (a few bytes). This result confirms the project's objective of "low-latency communication" and reduced "certificate administration overhead".<sup>1</sup>

### 5.3 System Usability and Interface Results

The visual results from the developed application demonstrate a functional and intuitive system.

- **Login Interface (Figure 5.1):** The secure entry point functions as intended, providing a barrier to unauthorized access while managing the retrieval of the sensitive private key.
- **Driver Dashboard (Figure 5.2):** This is the central hub of the application. It successfully displays:
  - **User Identity:** The user's License Plate (Public Key).
  - **Private Key:** The retrieved secret scalar (displayed for debugging/confirmation purposes in the report).
  - **Real-Time Alerts:** The "Incoming Alerts" section populates dynamically. The successful decryption of messages (e.g., "HEAVY TRAFFIC" appearing in red) serves as the ultimate proof-of-concept for the entire cryptographic pipeline.
- **Messaging Interface (Figure 5.3):** The logs generated by this interface verify that the application correctly identifies the target public key (Target License Plate) and creating an encrypted payload before transmission.

### 5.4 Data Management Logic Results

The backend results regarding data handling were also validated.

- **YAML Generation (Figure 5.7):** The automated generation of data.yaml files ensured that the YOLO training pipeline was reproducible and consistent.
- **Directory Creation (Figure 5.8):** The file system logic correctly organized the 1700 images into the appropriate folder hierarchy required by the neural network, preventing data leakage between training and testing sets.

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## Chapter 6: Discussion and Critical Analysis

## 6.1 Security Analysis of the Modified Architecture

The shift to a Diffie-Hellman-based PKG introduces specific security characteristics that merit discussion.

- **Resistance to Impersonation:** By binding the public key directly to the license plate, the system makes impersonation difficult. An attacker cannot simply generate a random key pair and claim it belongs to a specific vehicle; they must possess the private key  $d_{ID}$  that corresponds mathematically to the license plate string. Since only the PKG (possessing master secret  $s$ ) can generate  $d_{ID}$ , an attacker cannot forge a valid identity without compromising the central authority.
- **Forward Secrecy:** The use of ephemeral session keys (derived from the identity-based handshake) ensures that even if a long-term private key is compromised, past messages remain secure (provided the Diffie-Hellman exchange logic includes ephemeral components).
- **Key Escrow:** A feature (or drawback, depending on perspective) of this system is Key Escrow. The PKG knows the master secret  $s$  and can therefore derive *any* vehicle's private key. In a traffic management context, this is often a desired feature, allowing law enforcement to decrypt communications for forensic purposes (e.g., accident reconstruction).

## 6.2 Scalability and Smart City Integration

The removal of certificates allows this system to scale almost infinitely. The PKG does not need to maintain a directory of all active public keys; it simply derives the private key for a user upon request. This "stateless" property of the public key infrastructure is ideal for Smart Cities where millions of IoT devices and vehicles interact. The "low-latency" results observed in the evaluation phase suggest that this architecture can support the high-frequency beaconing required for autonomous platooning and intersection collision avoidance.

## 6.3 Comparison with PKI

| Feature      | Traditional PKI (Certificates) | Proposed Modified IBC (DH-based) |
|--------------|--------------------------------|----------------------------------|
| Trust Anchor | Certificate Authority (CA)     | Private Key Generator (PKG)      |
| Public Key   | Random String (in Cert)        | License Plate Number             |
| Validation   | Verify Cert Chain + CRL        | Math Verification (Hash          |



|            |                          |                              |
|------------|--------------------------|------------------------------|
|            | Check                    | operations)                  |
| Bandwidth  | High (Cert + Signature)  | Low (ID + Signature)         |
| Latency    | High (due to validation) | Low (Real-time)              |
| Revocation | Complex (CRLs, OCSP)     | Simple (Key Expiry/Rotation) |

The comparison clearly highlights the efficiency gains of the proposed methodology for the specific constraints of the vehicular environment.

## Chapter 7: Conclusion and Future Scope

### 7.1 Conclusion

This research report detailed the successful design and implementation of a secure V2V communication framework that eliminates the need for digital certificates. By creating a unique synthesis of **Deep Learning-based License Plate Recognition (LPR)** and a **Diffie-Hellman-based Identity-Based Cryptography (IBC)** scheme, the project addresses the critical bottlenecks of latency and scalability that plague traditional PKI systems.

The methodology demonstrated that a **Private Key Generator (PKG)** using modular exponentiation with a master secret ( $d = ID^s \pmod p$ ) can effectively issue keys based on visual identities extracted by **YOLOv8**. The system's multi-tiered architecture—spanning LPR, Cryptography, Communication, and Application layers—was validated through extensive testing.

- **Backend Testing** confirmed the robust management of user identities and secure key generation.
- **LPR Evaluation** proved the viability of using standard cameras to derive cryptographic public keys in real-time.
- **Mobile Testing** showcased the seamless integration of these technologies, providing drivers with a "dashboard" that decrypts safety alerts instantaneously.

The project successfully maintains the "essence" of the IBC framework—trust centralized in a PKG and identity derived from attributes—while optimizing the mathematical engine for the resource-constrained reality of vehicular networks.

### 7.2 Future Scope

To transition this research from a prototype to a deployment-ready solution, several avenues for future work are identified:

1. **Hardware Integration:** Implementing the cryptographic module on dedicated Hardware Security Modules (HSM) or Trusted Platform Modules (TPM) within vehicle ECUs to secure the storage of the private key.
2. **Advanced LPR Techniques:** Integrating Generative Adversarial Networks (GANs) to super-resolve images taken in adverse weather (fog, heavy rain) or low-light conditions, ensuring the "Public Key" can always be read.
3. **Threshold Cryptography:** distributing the Master Secret  $S$  among multiple authorities to prevent a single point of compromise at the PKG level.
4. **Field Trials:** Conducting large-scale field tests with actual vehicles to measure signal propagation, packet loss rates, and end-to-end latency in highway and urban canyon scenarios.

This research lays a solid foundation for the next generation of secure, efficient, and intelligent vehicular communication systems.

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(End of Report)

## Works cited

1. RG37\_7SReport.docx
2. Identity-based Encryption from the Diffie-Hellman Assumption, accessed on January 30, 2026, <https://cispa.de/en/research/publications/53312-identity-based-encryption-from-the-diffie-hellman-assumption>
3. Key Exchange in SSL/TLS: Understanding RSA, Diffie-Hellman, and Elliptic Curves - Tech Papers - Citrix Community, accessed on January 30, 2026, <https://community.citrix.com/tech-zone/build/tech-papers/key-exchange-in-ssl-tls/>
4. Identity-based Cryptography - Thales, accessed on January 30, 2026, <https://cpl.thalesgroup.com/blog/access-management/identity-based-cryptography>
5. Verifiable Security of Boneh-Franklin Identity-Based Encryption\* - The IMDEA Software Institute, accessed on January 30, 2026, <https://software.imdea.org/~szanella/Zanella.2011.ProvSec.pdf>
6. Boneh-Franklin IBE - A Few Thoughts on Cryptographic Engineering, accessed on January 30, 2026, <https://blog.cryptographyengineering.com/boneh-franklin-ibe/>
7. Identity-Based Encryption from the Weil Pairing - Applied Cryptography Group, accessed on January 30, 2026, <https://crypto.stanford.edu/~dabo/papers/bfibe.pdf>
8. 8.1 Ciphers using modular exponentiation | MATH1001 Introduction to Number

Theory, accessed on January 30, 2026,

<https://www.southampton.ac.uk/~wright/1001/ciphers-using-modular-exponentiation.html>

9. 11.3 A Modular Exponentiation Cipher - Mathematics and Computer Science, accessed on January 30, 2026,

<https://math.gordon.edu/ntic/ntic/section-diffie-hellman.html>